A Secure, Reliable and Low-Cost Distributed Storage Scheme Based on Blockchain and IPFS for Firefighting IoT Data

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by

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Certificate

This is to certify that the seminar report titled **A Secure, Reliable and Low-Cost Distributed Storage Scheme Based on Blockchain and IPFS for Firefighting IoT Data** is a bonafide record of the work carried out by **Muhammed Hafis (Univ.Reg.No. VAS23CS138)** of Vidya Academy of Science & Technology, Thalakkottukara, Thrissur - 680 501 in partial fulfillment of the requirements for the award of **Degree of Bachelor of Technology** in **Computer Science and Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2023-2027.

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Acknowledgement

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Abstract

In fire incident investigations, firefighting Internet of Things (IoT) data such as sensor feeds and 24-hour surveillance video is the primary electronic evidence used for event analysis, tracing the origin of a fire and determining legal responsibility. Centralized storage of this evidence is exposed to tampering by malicious insiders or hackers and to physical loss, because the fire itself can destroy the servers, while storing large videos directly on a blockchain forces every consortium node to duplicate the files and causes the ledger size to explode. This seminar studies a secure, reliable and low-cost distributed storage scheme that combines Hyperledger Fabric with the InterPlanetary File System (IPFS). The data is compressed, encrypted with a random AES key and stored off-chain in IPFS, which returns a content identifier (CID), whereas only the lightweight CID and metadata are recorded on the blockchain. The AES key and the CID are protected with RSA public-key encryption so that no plaintext is exposed. To protect against malicious nodes, a 5-phase Practical Byzantine Fault Tolerance (PBFT) consensus is embedded into the orderer of Hyperledger Fabric, which by default offers only Raft or Kafka. Experimental results show that the scheme reduces the peer ledger storage requirement by about 50%, achieves about 300 transactions per second under PBFT, compared with about 7 for Bitcoin and 15 for Ethereum, and finalizes transactions in 0.75 seconds on average, with AES encryption and decryption taking about 25 ms and RSA operations taking microseconds.

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Vision of the Institution

To be a premier institution in engineering education and research, nurturing innovative professionals for sustainable societal development.

Mission of the Institution

M1. Provide quality engineering education that empowers students with strong technical competence and analytical problem solving abilities.

M2. Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3. Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Vision of the Department

To excel in computer science and engineering education and research, nurturing innovative and ethically responsible professionals for sustainable technological development.

Mission of the Department

M1. Impart quality education in computer science and engineering by providing strong theoretical foundations, practical skills and exposure to emerging technologies.

M2. Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable and socially relevant computing solutions.

M3. Develop graduates with professional competence, ethical values, logical and analytical thinking skills, leadership qualities and a commitment to lifelong learning and societal well-being.

Program Educational Objectives (PEOs)

PEO-1: Graduates shall practice computer science and engineering in IT industry, public sector undertaking, as an entrepreneur or as a researcher pursuing a successful professional career.

PEO-2: Graduates shall demonstrate creativity, develop innovative ideas and work in teams to accomplish a common goal.

PEO-3: Graduates shall demonstrate project management skills, lifelong learning and social awareness upholding ethical values.

Program Outcomes (POs)

PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health, safety, and environment. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations using research-based knowledge including design of experiments, modelling, analysis and interpretation of data. (WK8)
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques and modern engineering tools recognizing their limitations. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze societal and environmental impacts with reference to sustainability, economy, health, safety, and culture. (WK1, WK5, WK7)
PO7	**Ethics:** Apply ethical principles, professional ethics, and adhere to laws. (WK9)
PO8	**Individual and Collaborative Team Work:** Function effectively as an individual and in teams.

PO9	**Communication:** Communicate effectively through reports, documentation, and presentations considering diverse audiences.
PO10	**Project Management and Finance:** Apply management principles and economic decision-making in projects.
PO11	**Life-Long Learning:** Recognize the need for independent and lifelong learning. (WK8)

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Specific Outcomes (PSOs)

PSO-1	The ability to apply the knowledge of computational structures and algorithms for solving complex engineering problems in multidisciplinary areas.
PSO-2	The ability to design, develop and evaluate computer-based systems using advanced tools and technologies to meet real-world challenges.

Sustainable Development Goals (SDGs)

SDG 1	**No Poverty:** End poverty in all its forms everywhere.
SDG 2	**Zero Hunger:** End hunger, achieve food security and improved nutrition and promote sustainable agriculture.
SDG 3	**Good Health and Well-being:** Ensure healthy lives and promote well-being for all at all ages.
SDG 4	**Quality Education:** Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
SDG 5	**Gender Equality:** Achieve gender equality and empower all women and girls.
SDG 6	**Clean Water and Sanitation:** Ensure availability and sustainable management of water and sanitation for all.
SDG 7	**Affordable and Clean Energy:** Ensure access to affordable, reliable, sustainable and modern energy for all.
SDG 8	**Decent Work and Economic Growth:** Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.
SDG 9	**Industry, Innovation and Infrastructure:** Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.
SDG 10	**Reduced Inequalities:** Reduce inequality within and among countries.
SDG 11	**Sustainable Cities and Communities:** Make cities and human settlements inclusive, safe, resilient and sustainable.
SDG 12	**Responsible Consumption and Production:** Ensure sustainable consumption and production patterns.

SDG 13	**Climate Action:** Take urgent action to combat climate change and its impacts.
SDG 14	**Life Below Water:** Conserve and sustainably use the oceans, seas and marine resources for sustainable development.
SDG 15	**Life on Land:** Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.
SDG 16	**Peace, Justice and Strong Institutions:** Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.
SDG 17	**Partnerships for the Goals:** Strengthen the means of implementation and revitalize the global partnership for sustainable development.

SDGs addressed with Justification

Seminar Title	**SDG addressed**	**Justification**
Distributed Storage for Firefighting IoT Data (Blockchain and IPFS)	SDG 9: Industry, Innovation and Infrastructure	Presents a secure, reliable and low-cost storage infrastructure for firefighting IoT evidence, using IPFS, Hyperledger Fabric and PBFT consensus, that survives the loss of a local server. This is the primary goal addressed.
	SDG 3: Good Health and Well-being	Tamper-resistant fire evidence supports accurate investigation of incidents and their causes, which helps improve fire safety and protect lives.

POs & PSOs addressed with Justification

Seminar Title	**POs & PSOs addressed**	**Justification**
Distributed Storage for Firefighting IoT Data (Blockchain and IPFS)	PO1 – Engineering Knowledge	Applies cryptography (AES, RSA), distributed storage, blockchain consensus (PBFT) and IPFS fundamentals to store fire evidence securely.
	PO2 – Problem Analysis	Analyzes the weaknesses of centralized evidence storage such as tampering, physical loss and ledger growth, and relates them to legal responsibility.
	PO3 – Design/Development of Solutions	Studies a design that keeps large video off-chain in IPFS and records only CIDs and metadata on Hyperledger Fabric to cut cost and protect evidence.
	PO4 – Conduct Investigations of Complex Problems	Reviews the literature on IPFS, Hyperledger Fabric and PBFT and interprets reported results on storage, throughput and latency.
	PO5 – Modern Tool Usage	Examines Hyperledger Fabric, IPFS, AES and RSA, and recognizes limits such as PBFT scalability and key management.
	PO6 – The Engineer and The World	Supports fire investigation and legal accountability, and reduces the loss of critical evidence after fire incidents.

Seminar Title	**POs & PSOs addressed**	**Justification**
	PO7 – Ethics	Keeps plaintext off the ledger through AES and RSA protection and records evidence access transparently.
	PO8 – Individual and Team Work	Works independently and incorporates feedback from the guide, coordinators and peers during the seminar.
	PO9 – Communication	Communicates the study through this report and a seminar presentation.
	PO10 – Project Management and Finance	Plans the seminar within the schedule and considers gas, hosting and infrastructure cost in the design.
	PO11 – Life-Long Learning	Studies emerging technologies independently, drawing on research papers and documentation.
	PSO1	Applies hashing, encryption and consensus rules to decide how fire data is stored and verified.
	PSO2	Evaluates a distributed storage scheme using modern tools against real evidence-storage needs.

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List of Abbreviations

IPFS	InterPlanetary File System
IoT	Internet of Things
CID	Content Identifier
AES	Advanced Encryption Standard
RSA	Rivest–Shamir–Adleman
PBFT	Practical Byzantine Fault Tolerance
BFT	Byzantine Fault Tolerance
DHT	Distributed Hash Table
DAG	Directed Acyclic Graph
TPS	Transactions Per Second
ZKP	Zero-Knowledge Proof
PK	Public Key

Chapter 1

INTRODUCTION

1.1 Background and Motivation

In fire incident investigations, firefighting Internet of Things (IoT) data is the primary electronic evidence required for event analysis, tracking the point of origin of a fire and determining legal responsibility. Modern fire scenes are monitored by many devices. Sensor feeds report water pressure, temperature, gas, smoke and vibration as very small JSON records of about 1 KB, while video monitoring produces 24-hour, 1080P multi-camera MP4 streams of about 50 GB. Since the outcome of an investigation may involve large financial and legal liability, the stored data must be trustworthy long after the incident.

Two requirements follow from this role. The first is *immutability*: electronic logs and network triggers must resist manual deletion or modification after the disaster. The second is *completeness*: the storage system must secure lightweight sensor data and massive bandwidth video streams seamlessly within the same framework.

Conventional centralized servers cannot guarantee either requirement. Malicious actors, internal employees or hackers can manipulate database logs to shift blame, the physical servers can be destroyed by the fire itself, and the history of data access is difficult to verify. These weaknesses motivate a decentralized, tamper-proof and scalable storage scheme for firefighting IoT data.

1.2 Technological Advancements in Distributed Storage

Blockchain technology provides a decentralized ledger in which records are linked cryptographically and cannot be altered without detection. Permissioned platforms such as Hyper-

ledger Fabric allow a consortium of organizations, for example fire departments, municipal bodies, insurers and courts, to maintain a shared ledger with identity-controlled access. Their drawback is that a blockchain is a poor place to store large files: a 50 GB surveillance video stored in a typical blockchain forces every consortium node to keep a duplicate copy, the ledger grows rapidly, and the large payloads reduce the transaction throughput.

The InterPlanetary File System (IPFS) offers a content-addressed, peer-to-peer storage network in which files are split into blocks, organized as a Merkle DAG and located by the hash of their content. It scales well for large data, but it provides no access control on its own, so confidentiality must be added by encrypting the data before it is stored.

Another important advancement is Byzantine fault tolerant consensus. The official Hyperledger Fabric (v1.4 and later) ordering services use Raft or Kafka, which perform very well but tolerate only crash failures. For evidence data that can be a target of attack, a Byzantine fault tolerant protocol such as Practical Byzantine Fault Tolerance (PBFT) is needed.

1.3 Role of Blockchain and IPFS in Firefighting Data Storage

The studied scheme assigns complementary roles to the two technologies. The complete (encrypted) data resides off-chain in IPFS, which returns a content identifier (CID) of 46 characters that cryptographically represents the content. Only this lightweight CID and associated metadata are written to the Hyperledger Fabric ledger. The ledger therefore stays small and fast, while the immutable record on the chain proves that the stored evidence has not been changed, because any modification of the data would change its CID.

Public-key and symmetric cryptography complete the design: AES protects the bulk data, and RSA protects the AES key and the address hashes, so that raw blocks on IPFS nodes cannot be read by eavesdroppers.

1.4 Goals and Purposes

The overarching goal of this study is to present a secure, reliable and low-cost distributed storage scheme for firefighting IoT data, which preserves the integrity of fire-scene evidence and remains practical for large video payloads.

A second goal is to remove the storage scale dilemma by means of a hybrid on-chain and off-chain model, so that the peer ledger storage requirement is substantially reduced. A third goal is to provide Byzantine fault tolerance by embedding a 5-phase PBFT consensus into the Hyperledger Fabric orderer, without losing the practical transaction speed that is required in real deployments.

Finally, the study evaluates the scheme in terms of storage space, throughput and latency, and outlines how it can be extended to multi-organization governance, fine-grained access control and privacy-preserving technologies.

Chapter 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

2.1.1 IPFS - Content Addressed, Versioned, P2P File System

- **Authors:** Benet, 2014
- **Source:** arXiv preprint (arXiv:1407.3561)

Benet (2014) introduced IPFS as a peer-to-peer distributed file system that combines a distributed hash table, an incentivized block exchange and a self-certifying namespace. Files are divided into blocks organized as a Merkle directed acyclic graph, and every object is addressed by the hash of its content, which gives built-in integrity verification, de-duplication and efficient distribution.

IPFS is cost-efficient and scalable for large files such as surveillance video. However, any peer that knows a hash can retrieve the corresponding data, and IPFS has no native encryption or access control. Sensitive evidence therefore has to be encrypted before it is added to IPFS, which is the approach followed in the studied scheme.

2.2 LITERATURE SURVEY

2.2.1 Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains

- **Authors:** Androulaki et al., 2018
- **Source:** Proceedings of the Thirteenth EuroSys Conference, ACM

Androulaki et al. (2018) presented Hyperledger Fabric, a modular permissioned blockchain platform that uses an execute-order-validate architecture. Smart contracts, called chaincode, are executed by endorsing peers, an ordering service establishes the total order of transactions, and peers validate and commit them to the ledger. Membership is controlled through identities, which makes the platform suitable for consortia of known organizations.

The ordering service of Fabric is pluggable, and the official releases provide crash fault tolerant options such as Raft and Kafka. These offer high throughput but no protection against malicious (Byzantine) ordering nodes, which is the limitation addressed by embedding PBFT in the studied scheme.

2.3 LITERATURE SURVEY

2.3.1 Practical Byzantine Fault Tolerance

- **Authors:** Castro and Liskov, 1999
- **Source:** Symposium on Operating Systems Design and Implementation (OSDI), USENIX

Castro and Liskov (1999) proposed PBFT, a state machine replication algorithm that tolerates Byzantine faults in asynchronous networks and is efficient enough for practical use. With $3f + 1$ replicas the system tolerates up to f faulty nodes. The protocol proceeds through a request, a pre-prepare phase in which the primary proposes an ordering, a prepare phase in which replicas agree on that ordering, a commit phase and a reply to the client.

PBFT provides strong safety for permissioned settings, but its message complexity limits its scalability to a moderate number of nodes. This makes it appropriate for a consortium blockchain of a limited number of organizations, as in the firefighting evidence storage scenario.

Chapter 3

COMPONENTS OF THIS SYSTEM

3.1 Firefighting IoT Data Sources and Local Terminal

The data to be protected comes from two kinds of sources. Sensor feeds are small JSON records of about 1 KB reporting water pressure, temperature, gas, smoke and vibration. Video monitoring consists of 24-hour 1080P multi-camera streams of about 50 GB in MP4 format. Both are collected by a local terminal, which compresses the sensor logs and videos into a ZIP archive, generates a random symmetric key and performs the encryption before anything leaves the premises.

3.2 IPFS Off-Chain Storage

IPFS stores the complete encrypted archive off-chain. The file is chunked into blocks of 256 KB, organized as a Merkle DAG, and the root hash is returned as the content identifier (CID) of 46 characters. Retrieval is performed through DHT addressing, and the Merkle structure allows the integrity of the downloaded data to be verified automatically.

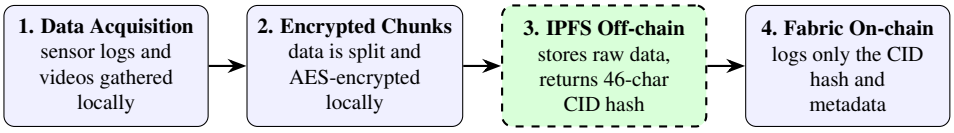


Figure 3.1: Hybrid off-chain and on-chain storage model

3.3 Hyperledger Fabric with Embedded PBFT

Hyperledger Fabric acts as the on-chain component. It stores only the CID and the metadata of every upload and enforces access through its chaincode. Official Fabric (v1.4 and later) uses Raft or Kafka, which lack Byzantine fault tolerance and therefore remain vulnerable to malicious attackers. In the studied scheme, a 5-phase PBFT algorithm is compiled and embedded directly into the orderer package.

3.4 Cryptographic Dual-Protection System

To avoid eavesdropping or raw block leakage on IPFS nodes, a dual-layer cryptographic design is adopted. **AES** (symmetric) is highly parallel and uses little memory, so it is applied directly to the raw firefighting archives with their huge video payloads. **RSA** (asymmetric, 1024/2048-bit) is used for the secure distribution of the single-use AES keys and the IPFS address hashes.

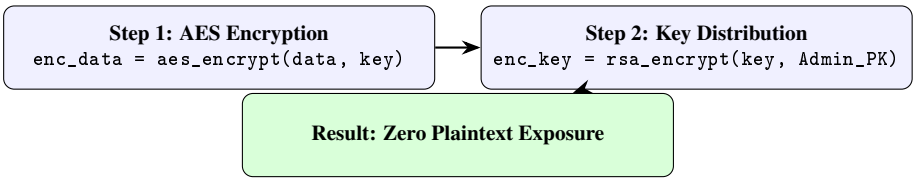


Figure 3.2: Encryption pipeline

Chapter 4

METHODOLOGY

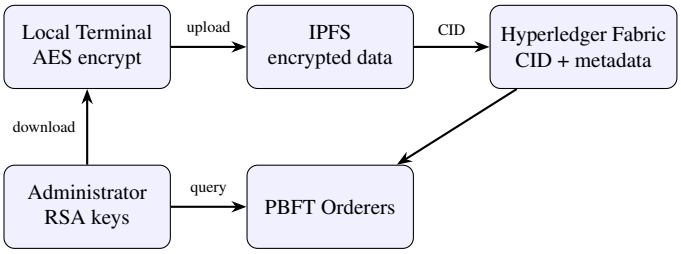


Figure 4.1: Working flow chart

4.1 Working Mechanism

4.1.1 Working Mechanism

The proposed scheme resolves the storage scale dilemma through a **Hybrid On-Chain/Off-Chain Distributed Model**. The complete data resides securely in IPFS, while only lightweight cryptographic addresses populate the Hyperledger Fabric ledger. Data is encrypted locally, so neither the IPFS nodes nor the blockchain peers ever see plaintext, and the blockchain provides an immutable and auditable record of every stored item.

4.1.2 Data Upload Workflow

The upload proceeds in four steps. (1) *AES encryption:* the local terminal generates a random symmetric key and encrypts the ZIP archive containing the sensor data and video. (2) *Key seeding:* the symmetric key is encrypted with the administrator's RSA public key and

transferred securely. (3) *IPFS pinning:* the file is chunked (256 KB) in IPFS, which generates the Merkle DAG root CID hash. (4) *On-chain submission:* the hash is signed with the user's private key and pushed to the Hyperledger ledger.

4.1.3 Query and Verification Workflow

Retrieval also has four steps. (1) *Hash access:* an authorized administrator requests the on-chain metadata through public key authentication. (2) *Decentralized download:* the administrator extracts the encrypted IPFS hash and requests the content from IPFS using DHT addressing. (3) *Symmetric decryption:* the administrator's private RSA key decrypts the encrypted AES key, which unlocks the original telemetry and video file. (4) *Validation:* the integrity of the data is verified automatically by Merkle tree matching.

4.1.4 Embedding PBFT Consensus in Hyperledger Fabric

Official Hyperledger Fabric (v1.4+) uses Raft or Kafka, which are high-performing but lack Byzantine fault tolerance. The 5-phase PBFT algorithm is compiled and embedded directly into the orderer package. In the *Request* phase the client sends the fire-log transaction proposal. In *Pre-Prepare* the primary orderer validates and broadcasts the proposal. In *Prepare* the nodes execute the audit validation and $2f+1$ agreement is needed. In *Commit* the leader confirms the local ledger write and broadcasts the state. In *Reply* the web application commits the transaction upon $f+1$ replies.

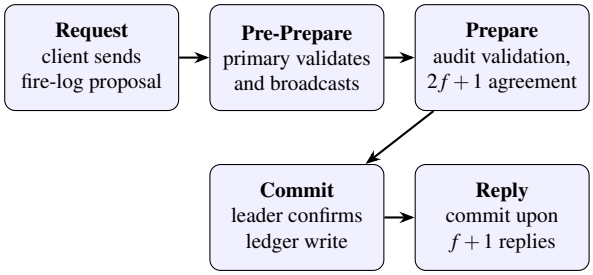


Figure 4.2: Five phases of PBFT embedded in the Fabric orderer

4.1.5 Smart Contract (Chaincode) Design

Two chaincode algorithms manage the ledger. **Algorithm 3,** `sendhash` **(uplink),** automates the creation of the secure ledger metadata record for a newly stored archive. **Algorithm 4,** `query` **(retrieval),** enforces the signature check of the requesting administrator before the metadata is released.

4.2 Results and Evaluation

4.2.1 Storage Space Consumption

Physical disk space was compared for 10 concurrent organizational nodes uploading high-definition stream blocks. In Scheme 1 (no blockchain) the space is minimal, about 62 MB in total, but the data is vulnerable to local physical deletion or loss. In Scheme 2 (traditional full on-chain storage) over 250 MB is used, because the complete video logs are repeated across four peer databases. In Scheme 3 (the proposed IPFS and blockchain scheme) the ledger size is reduced by about 50%, since only a CID text of about 1 KB is logged on-chain.

4.2.2 Blockchain Throughput

Stress tests measured transactions per second (TPS) with the official Hyperledger test suites at varying concurrencies. The default Raft/Kafka ordering reaches about 750 TPS but has no protection against Byzantine attacks. The proposed PBFT ordering achieves about 300 TPS while providing robust security, which outperforms public chains such as Bitcoin (about 7 TPS) and Ethereum (about 15 TPS).

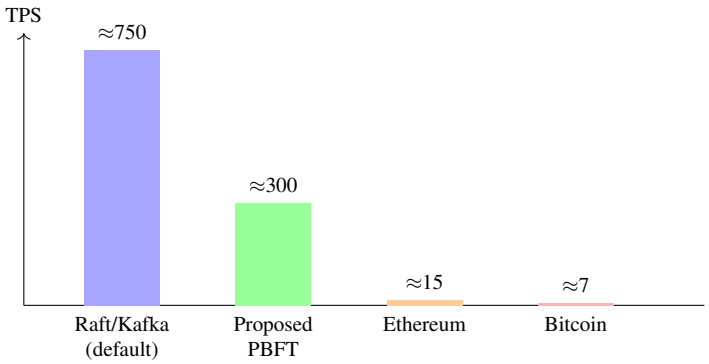


Figure 4.3: Maximum transaction throughput comparison

4.2.3 Time Latency and Cryptographic Overhead

The latency of every functional pipeline was tested to ensure usability; transactions are finalized in under 1 second on average. Table 4.1 lists the measured costs.

Table 4.1: Time latency and cryptographic overhead

Functional Process	Protocol	Average Time	Real-world Impact
Symmetric Media Encryption	AES-128 / 256	24.98 ms	Instantaneous compression and upload flow
Symmetric Media Decryption	AES-128 / 256	24.78 ms	Low lag file extraction at terminal
Asymmetric Hash Encryption	RSA-2048 Public Key	57.50 μs	Negligible impact on gateway latency
Asymmetric Hash Decryption	RSA-2048 Private Key	417.12 μs	Sub-millisecond verification time
Average PBFT Block Latency	Hyperledger Caliper	0.75 s	Concludes faster than public chain targets

Chapter 5

APPLICATION

5.1 Fire Incident Investigation:

The scheme preserves sensor logs and surveillance video of a fire scene as tamper-proof evidence, which supports event analysis and tracking of the origin point of the fire.

5.2 Legal Responsibility Determination:

Because the on-chain CID record is immutable and its history of access can be audited, courts and insurers can rely on the evidence when determining legal responsibility.

5.3 Smart Building and Industrial Safety Monitoring:

The same hybrid design can store continuous data from sensors and cameras in buildings, factories and warehouses where large media must be kept securely for long periods.

5.4 Disaster-Resilient Data Backup:

Since the data is distributed over IPFS nodes rather than a single server, the destruction of local equipment by fire does not erase the historical sensor data.

5.5 Multi-Organization Evidence Sharing:

A consortium of fire departments, municipal bodies, insurance companies and courts can share one permissioned ledger, each retrieving evidence through signed requests.

5.6 Low-Cost Large Media Archiving:

Keeping only a CID of about 1 KB on the chain reduces peer storage by about 50% and allows even small edge nodes to participate in the network.

5.7 Other Evidence-Critical IoT Domains:

The model is applicable wherever large IoT data must be stored with integrity guarantees, for example traffic accident recording and industrial incident logging.

Chapter 6

CHALLENGES AND LIMITATIONS

6.1 PBFT Scalability and Throughput

PBFT reduces the throughput from about 750 TPS (Raft/Kafka) to about 300 TPS, and its message overhead limits the number of ordering nodes, so the consortium must remain of moderate size.

6.2 Key Management Complexity

Single-use AES keys and RSA key pairs must be generated, stored and distributed securely. Loss of the administrator's private key would make the encrypted evidence unrecoverable, and key compromise would expose it.

6.3 Dependence on IPFS Availability

Data in IPFS remains available only while at least one node pins it. Reliable pinning, replication and monitoring are required so that evidence is not lost.

6.4 Handling Very Large Video Files

Uploading and encrypting archives of about 50 GB requires considerable bandwidth and processing at the local terminal, and retrieval time depends on the number and location of IPFS peers.

6.5 Privacy of Captured Footage

Video evidence may contain private details of victims. Currently authorized administrators can decrypt the whole archive, so privacy-preserving mechanisms are needed.

6.6 Integration and Implementation Effort

Embedding PBFT into the Fabric orderer requires modifying and compiling the platform code, which complicates upgrades, maintenance and compatibility with future Fabric releases.

6.7 Cost and Resource Requirements

Running IPFS nodes, Fabric peers and orderers requires infrastructure and operational expertise, even though the storage cost is lower than that of a full on-chain design.

Chapter 7

FUTURE SCOPE AND DISCUSSION

7.1 Future Developments

7.1.1 Multi-Organization Governance

The role structure can be standardized to include government, municipal, insurance and court entities, so that each organization has a defined part in storing and auditing fire evidence.

7.1.2 Fine-Grained Access Control

Role-based, fine-grained access policies can be implemented in chaincode so that different parties can access only the data they are entitled to.

7.1.3 Privacy-Preserving Technology

Zero-knowledge proofs (ZKP) can be integrated to selectively mask private, video-captured details of victims while still proving the authenticity of the evidence.

7.1.4 Performance Optimization of PBFT

Further optimization of the embedded PBFT consensus can narrow the throughput gap to Raft/Kafka and extend the scheme to larger networks.

7.1.5 Redundancy and Pinning Services

Replicated pinning across several organizations can increase the reliability of off-chain data.

7.1.6 Real-Time Streaming Support

Incremental chunking and anchoring would allow live video streams to be protected while the incident is still in progress.

7.1.7 Edge and IoT Integration

Lightweight encryption modules for edge devices can bring the protection closer to the data source.

7.1.8 Upgrade to Newer Fabric Releases

Porting the PBFT implementation to newer Hyperledger Fabric versions will keep the scheme maintainable.

7.1.9 Standardization and Collaborative Platforms

Common data formats and shared platforms for fire evidence would ease cooperation among agencies.

Chapter 8

CONCLUSION

The present study demonstrates a secure, reliable and low-cost distributed storage scheme for firefighting IoT data based on Hyperledger Fabric and IPFS. For the first time in this application, Hyperledger Fabric is combined with off-chain IPFS storage to protect the massive firefighting media and sensor data, with AES and RSA encryption ensuring that no plaintext is exposed.

The hybrid model reduces the local peer ledger storage requirement by about 50%. The embedded PBFT consensus provides Byzantine fault tolerance in Fabric 1.4 for increased security while still achieving about 300 TPS, which is well above Bitcoin and Ethereum, and transactions are finalized in less than one second with low cryptographic overhead.

Future work will extend the scheme with multi-organization governance, role-based access control and zero-knowledge proofs to protect the privacy of victims, thereby supporting trustworthy, evidence-grade storage of firefighting data.

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[7] *[Authors, venue and year of the base paper on firefighting IoT data storage using Blockchain and IPFS - fill from your reference paper].*

Appendix

- Reference Paper of the Seminar
  [Title of the base paper: A Secure, Reliable and Low-Cost Distributed Storage Scheme Based on Blockchain and IPFS for Firefighting IoT Data - add authors, venue and year]
- PPT of the Seminar



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